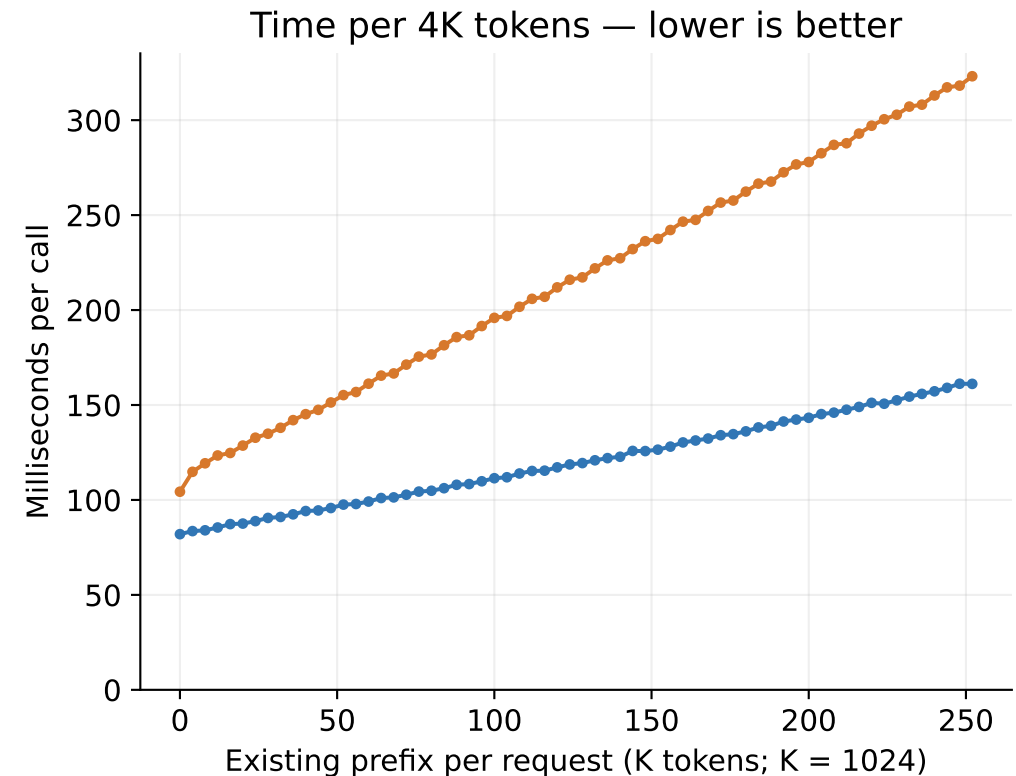
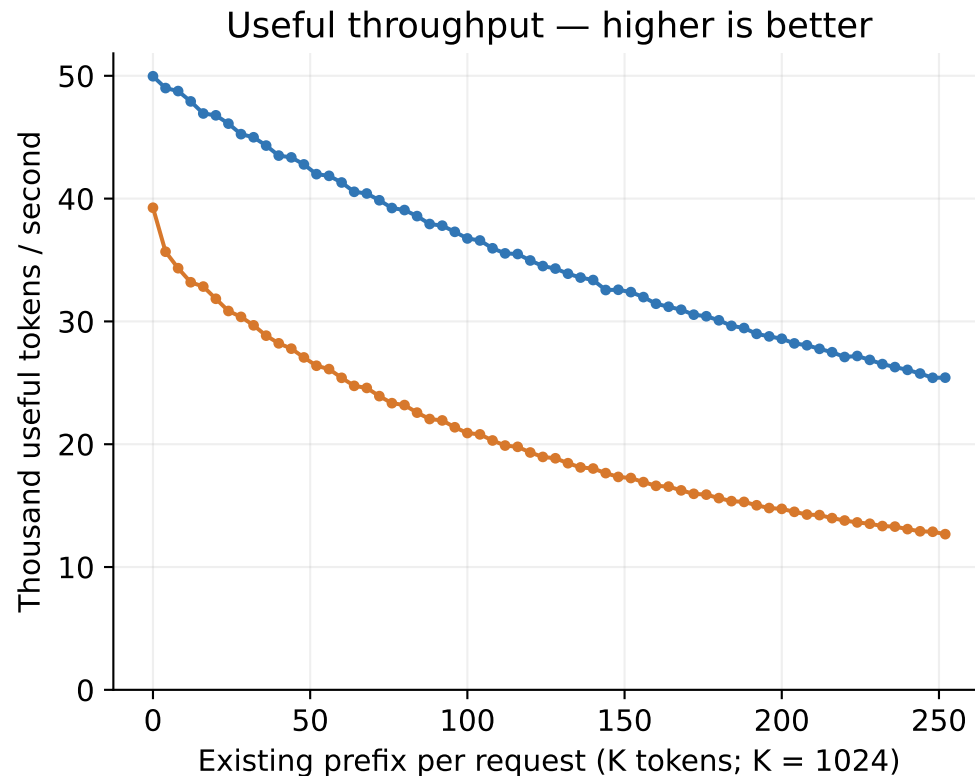


Fixed 4×1K batching vs canonical 4K

Same token count per call: 4,096 useful tokens. One request × 4K versus four requests × 1K.

— Canonical: 1 request × 4K — Fixed batch: 4 requests × 1K



0K prefix: canonical 82.0 ms; fixed batch 104.3 ms. The batch takes 27% longer for the same token count.

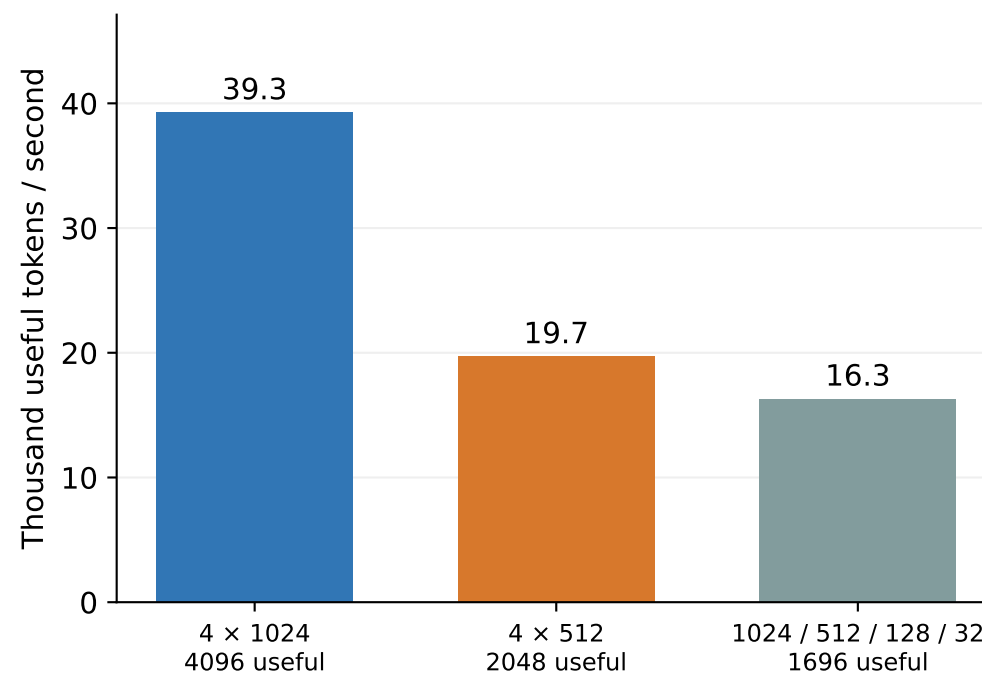
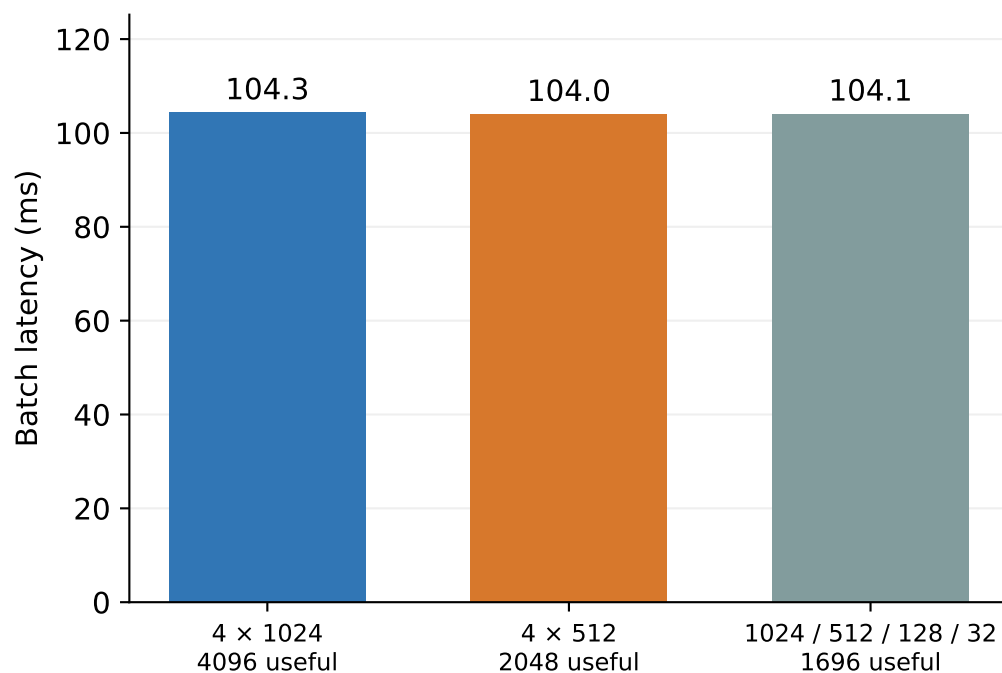
128K prefix: canonical 119.4 ms; fixed batch 217.2 ms. The batch takes 82% longer for the same token count.

248K prefix: canonical 161.2 ms; fixed batch 318.2 ms. The batch takes 97% longer for the same token count.

Gemma-4-31B-it, 60 layers, Blackhole CP8/TP4, original reduce-scatter, default activation placement. Each call processes 4,096 useful tokens: one 4K request in canonical, or four 1K requests in the fixed batch. Every request has the prefix shown on the chart. All histories were populated by real model calls. Five replays at prefixes 0, 8K, 32K, 64K, 128K, 192K and 248K; one replay at other positions. Timing covers trace replay and synchronization. Host staging is recorded separately; loading, compilation, output download and KV migration are excluded.

What does fixed-size padding cost?

All cases execute the same 4×1K shape. Short final requests still occupy their full 1K lane.



Why measure this? A fixed batch always computes 1K rows per request when fewer tokens are useful. All three cases start at zero and reuse the same trace.

The model shares embedding, projections, norms, RoPE and MLP across four requests. Each attention layer runs four sequential 1K attention calls with local slices and one local concatenation. It adds no CP all-gathers.

Memory: four 256K request caches fit. KV storage falls from 28.2 to 14.1 GiB/device when reducing eight slots to four. Weights and other buffers are additional. The canonical reference allocates one slot; both use full-history caches.

Validation: 132 host checks and six-layer output/KV, isolation, migration and trace replay checks passed. The strict 60-layer numerical comparison showed drift; further rounding analysis was stopped. Original math settings retained.